

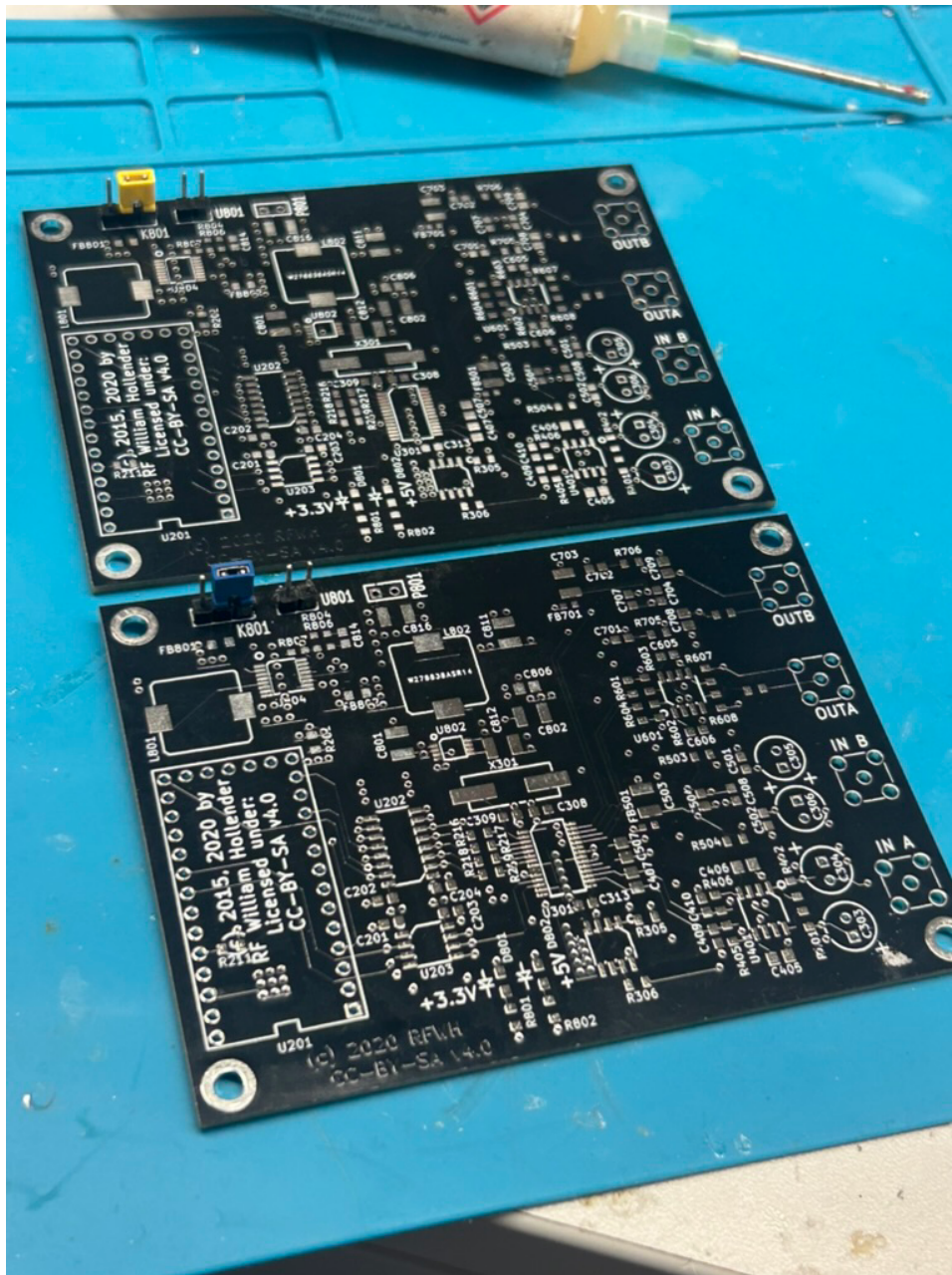
SuperAudioBoard illustrated assembly guide

This is a brief step-by-step guide to assembling and testing SuperAudioBoard, version for Teensy 4.0

Bill of materials: [Github URL](#)

Components cost per board = below 100 € (in Aug 2026)

Step 1. Mount headers for jumper K801 and U801



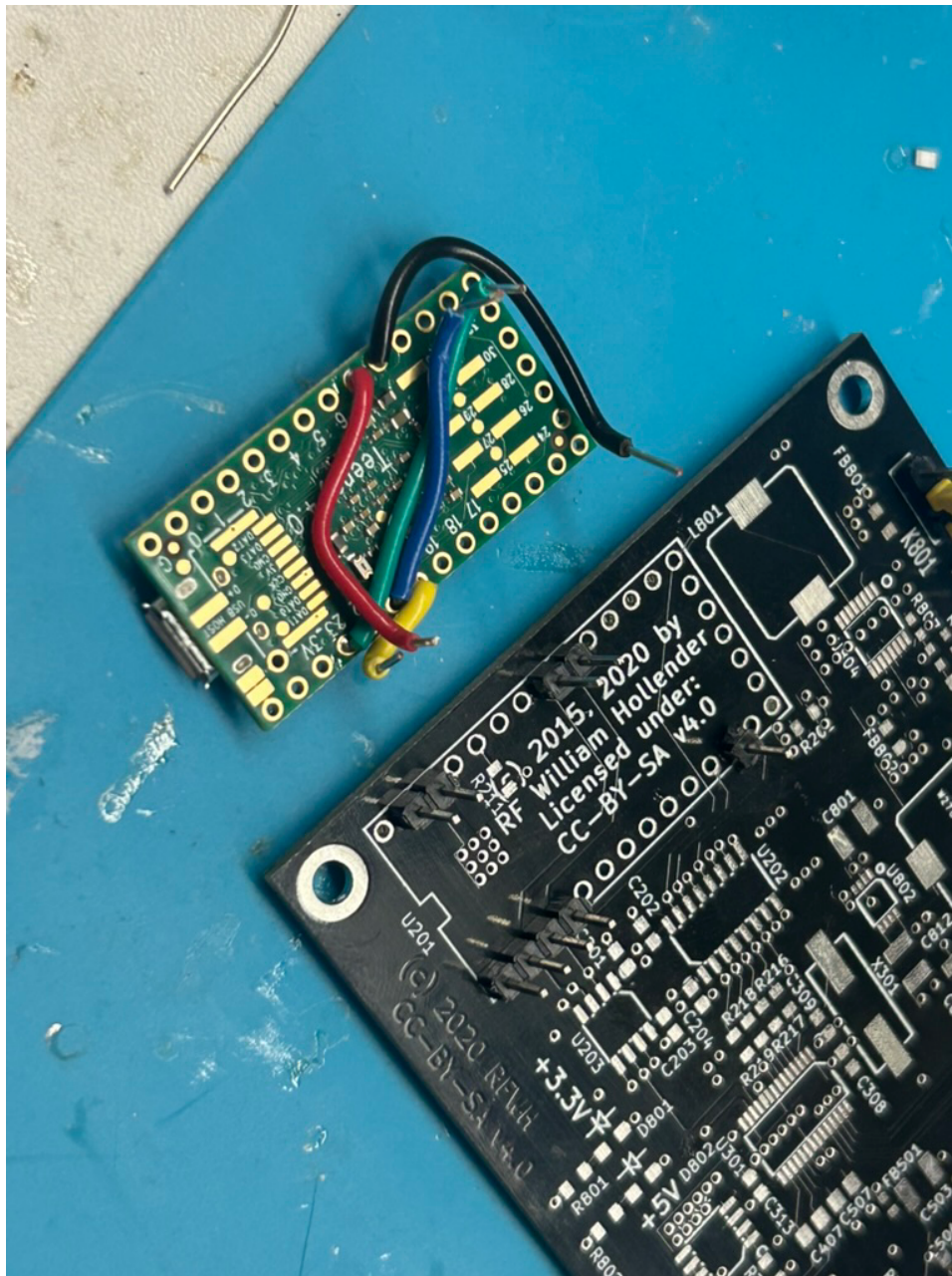
Step 2. Adapt headers for Teensy 4.0.

The PCB, unfortunately, still keeps the old pinout footprint of Teensy 3.x. For Teensy 4.0 we need to add some wires to reroute I²S connections. See

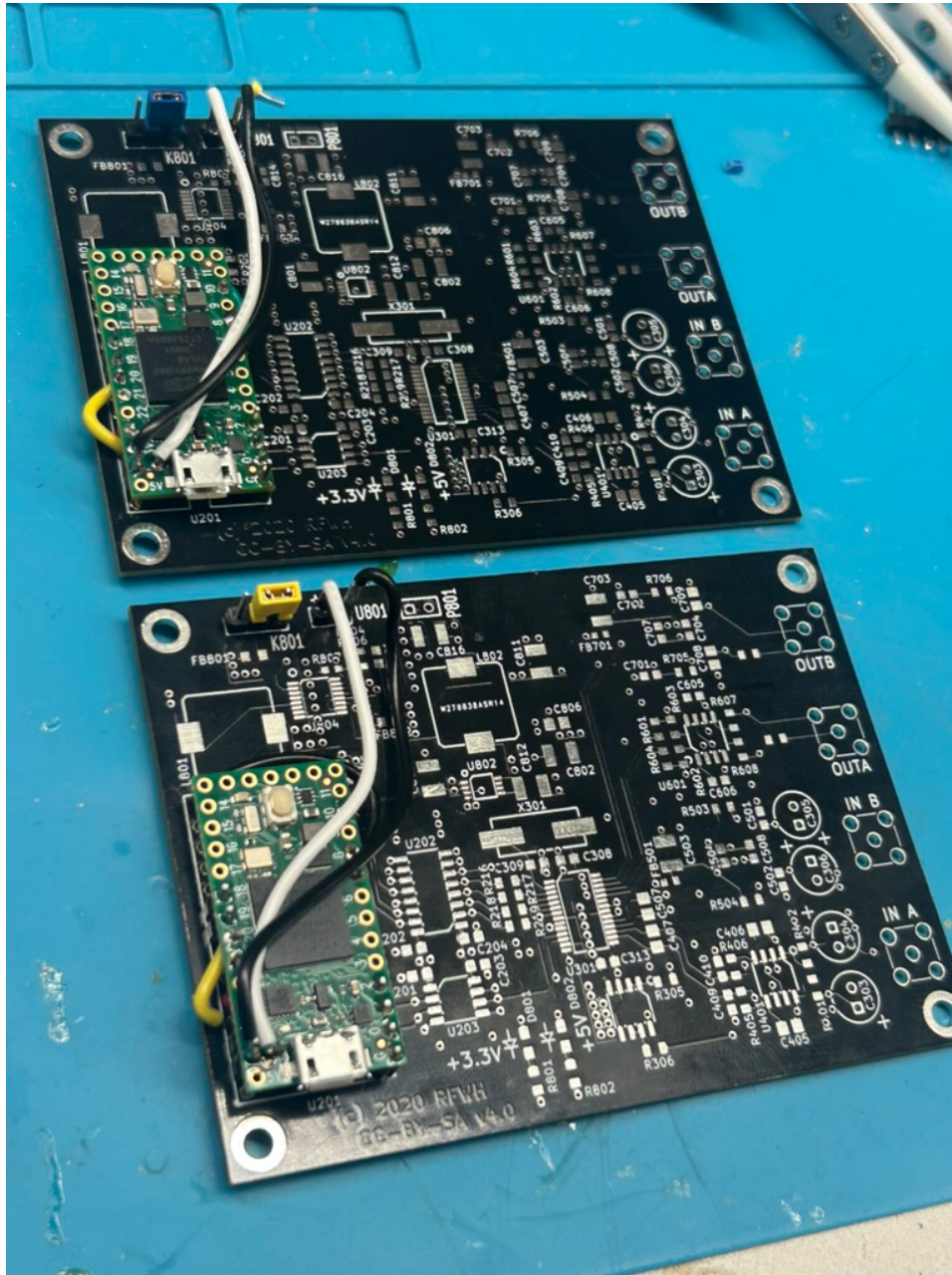
https://github.com/MichaelTayler/SuperAudioBoard/blob/main/Teensy_4.0/Pinoutdifferences.png.

Wire color	Teensy 3.2 pin	Teensy 4.0 pin	Pin function
Red	22	7	TX (in)
Black	13	8	RX (out)
Yellow	23	20	LRCLK1
Blue	9	21	BCLK1
Green	11	23	MCLK1

2A. Before soldering the Teensy, remember to add R211 (52 ohm SMD resistor)

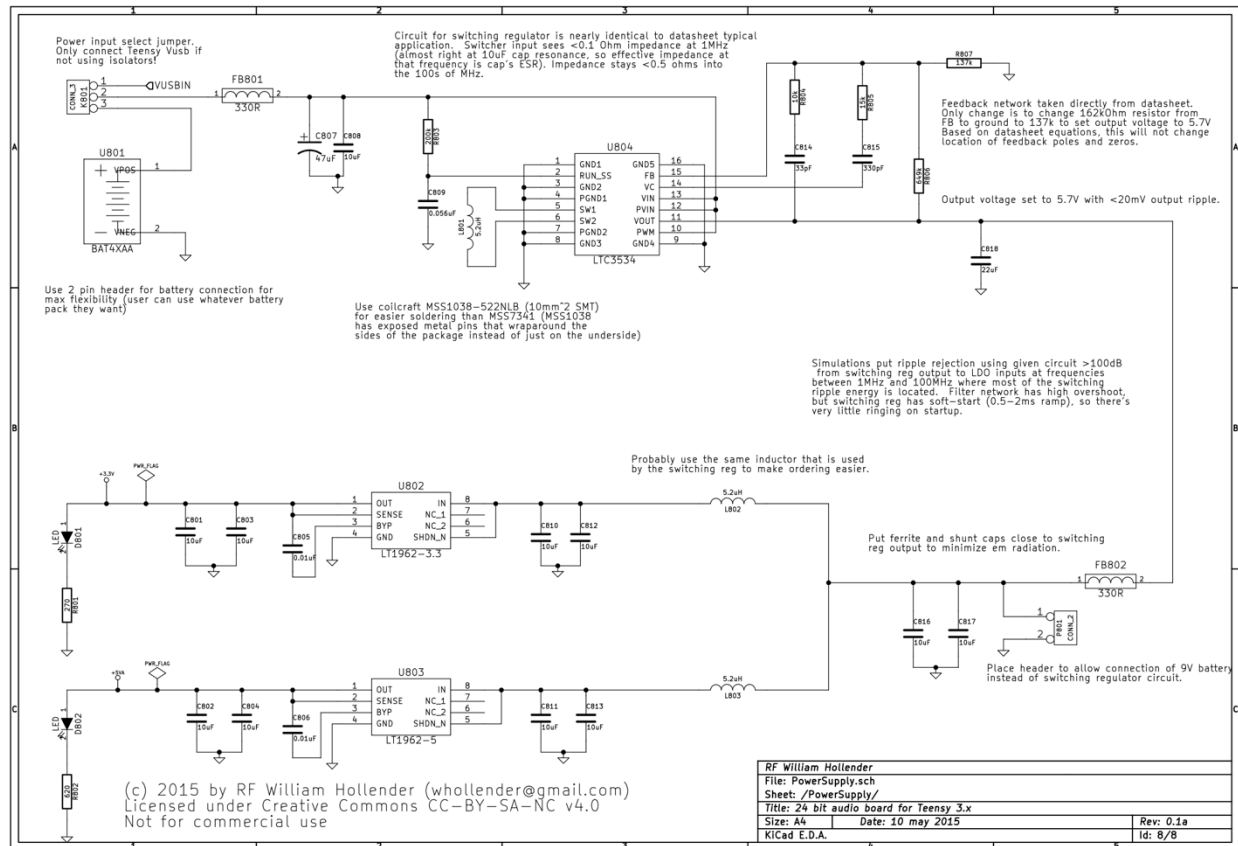


2B. Next, add wires between 5 V USB (white), GND (black) and header U801



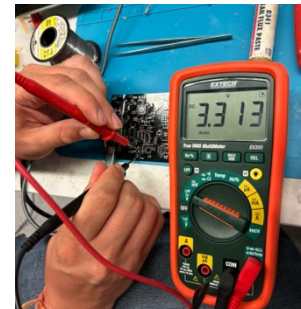
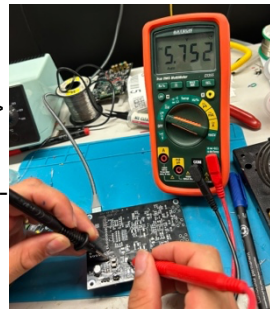
Step 3. VUSB power supply filter

We follow page 8 of the schematic, working clockwise from top left around to bottom left.
BAT4XAA and P801 are not used.

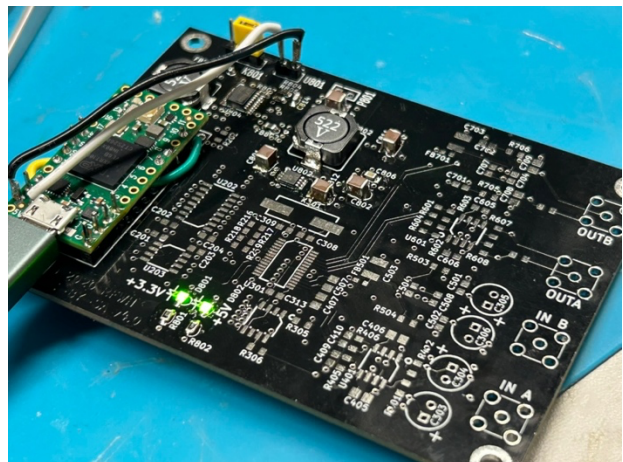


Test points:

- Between C808 and C809. V = 5.1 V
- Between C816 and L802. V = 5.75 V ---->
- Between C801 and D801. V = 5.08 V
- Between C802 and D802. V = 3.31 V ----->

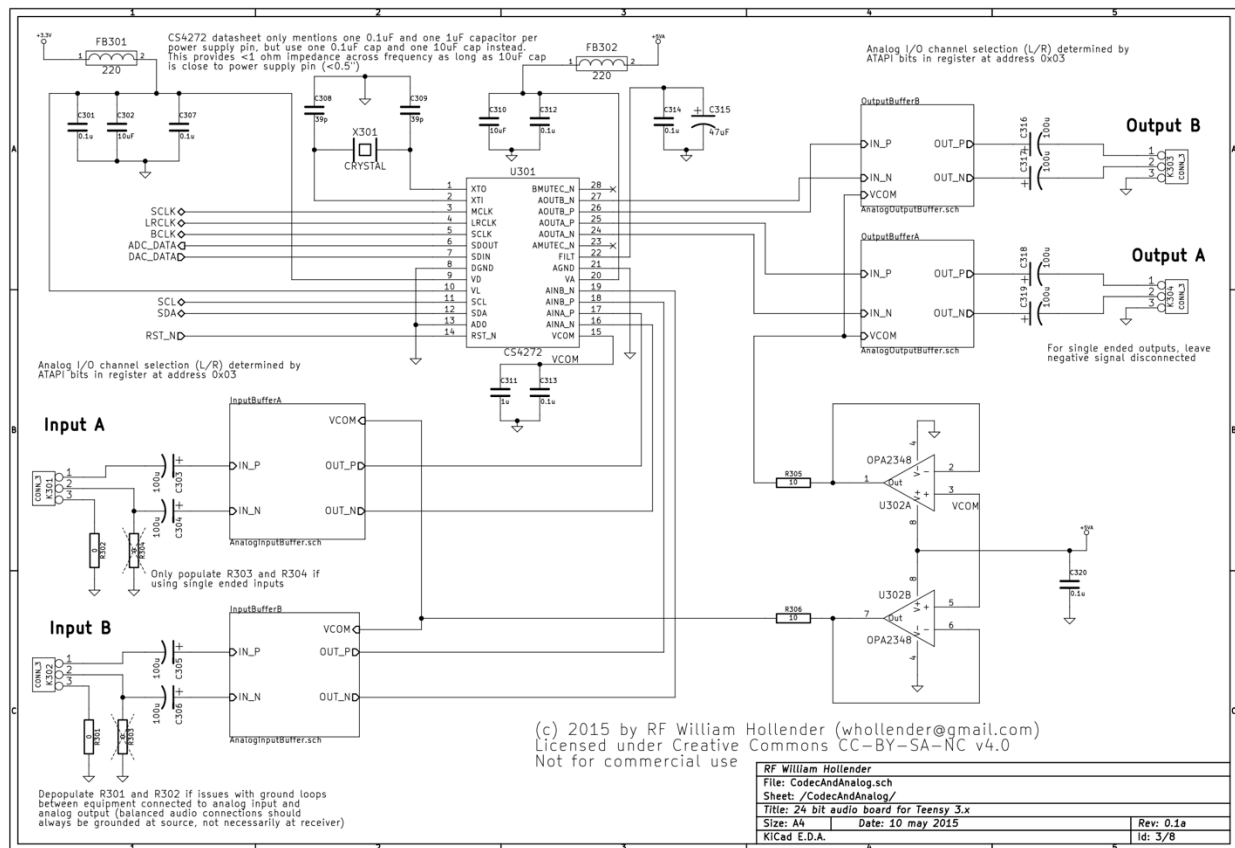


This step is complete once all voltages read OK, and LEDs D801 and D802 illuminate.

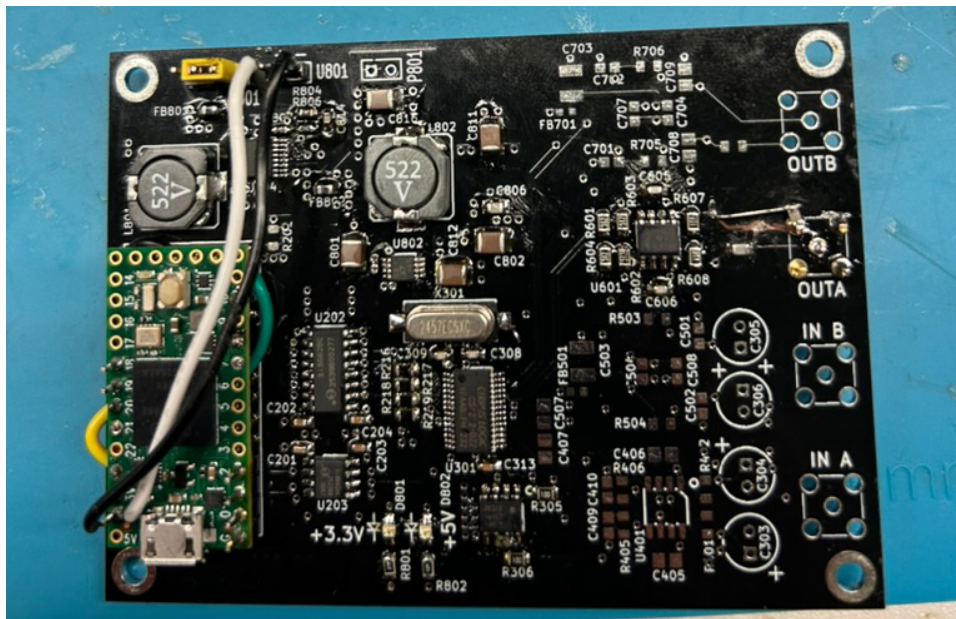


Step 4. Single channel output test

Use pages 2+3+6 of the [schematic](#) to populate the codec, logic isolator and OUTA output channel components. Soldering must be tidy, else bridges (especially on the small-pitch ICs) can occur leading to a non-working board. It is recommended to inspect the board under a microscope.



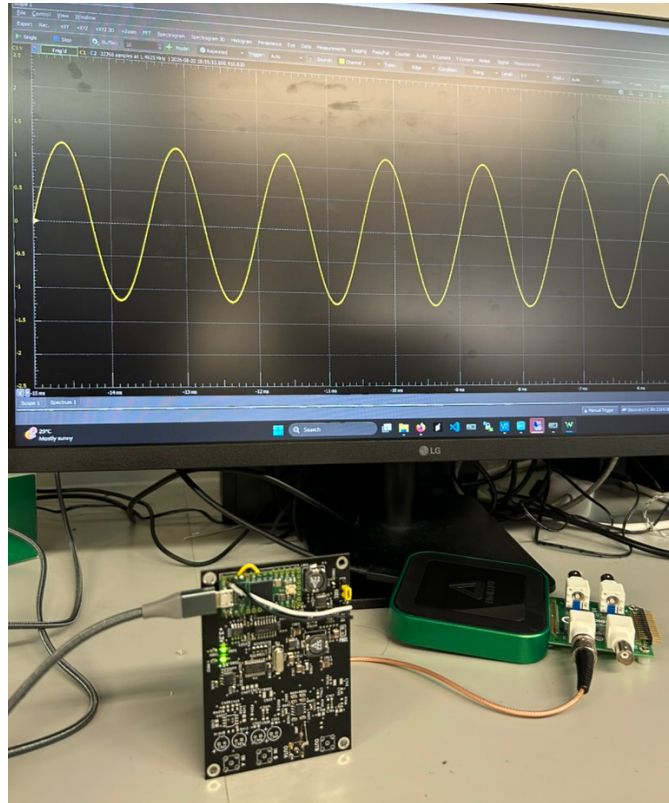
Board assembled up to sheets 1,2,3,6+8 of the schematic:



At this point, if not done already, connect the Teensy to a computer via USB and flash the test program. [Download from Github here.](#)

A small modification is made immediately before the OUTA SMA connector. The output is AC-coupled by adding a 10 nF capacitor in series. A suitable capacitor pad is already present on the board, but the PCB trace must be cut at the second connection point.

The output is tested using an Analog Discovery USB oscilloscope, though any modern oscilloscope should work. ----->



Step 5. To be added

To be added